

常微分方程第一周课程作业

4.24

一阶常微分方程

1. $y' = (1 - 2x)/y, y(1) = -2;$

2. $y' = x \sin x^2;$

3. $y + (2xy - e^{-2y})y' = 0;$

4. $ty' - y = t^2 e^{-t}, t > 0;$

5. $(2x - y) + (2y - x)y' = 0.$

6. $y' = \frac{xy}{x^2 + y^2};$

7. $y' = \frac{x}{x^2 + y^2}.$

4.28

1. $y^{(6)} + y = 0;$

2. $y^{(4)} - 4y''' + 4y'' = 0;$

3. $y^{(4)} + 2y''' + 2y'' = 3e^t + 2te^{-t} + e^{-t} \sin t;$

4. $y^{(6)} + y''' = t;$

5. $y''' - y'' + y' - y = g(t)$ (解用积分的形式表示);

$$1. y' = \frac{1-2x}{y}, y(1) = -2.$$

解: 变量分离, $ydy = (1-2x)dx$,
 积分 $\int_{-2}^y sds = \int_1^x (1-2t)dt$
 解得 $y = \sqrt{-2x^2 + 2x + 4}$

□

$$2. y' = x \sin x^2$$

解: 直接积分 $y = -\frac{1}{2} \cos x^2 + C$

□

$$3. y + (2xy - e^{-2y})y' = 0$$

解: $ydx + (2xy - e^{-2y})dy = 0$

$$\frac{\partial M}{\partial y} = 1, \frac{\partial N}{\partial x} = 2y.$$

$$-\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = 2 - \frac{1}{y}$$

故方程存在只与 y 有关的积分因子 $\mu(y) = e^{\int 2 - \frac{1}{y} dy} = \frac{e^{2y}}{y}$

两边同乘积分因子 $e^{2y}dx + (2xe^{2y} - \frac{1}{y})dy = 0$.

这是 $xe^{2y} - \ln y$ 的全微分, 故解为 $xe^{2y} - \ln y = C$

□

$$4. ty' - y = t^2 e^{-t}$$

解: 变形为 $y' = \frac{y}{t} + te^{-t}$.

先考虑齐次方程 $y' = \frac{y}{t}$

解得 $y = ct$

常数变易法 $y = c(t)t$ 带入原方程,

$$c'(t) = e^{-t}$$

$$c(t) = -e^{-t} + C$$

故解为 $y = -te^{-t} + Ct$.

□

$$5. (2x - y) + (2y - x)y' = 0$$

解: 变形为 $(2x - y)dx + (2y - x)dy = 0$.

$$\frac{\partial M}{\partial y} = -1 = \frac{\partial N}{\partial x} = -1.$$

故这是恰当方程.

注意到这是 $x^2 - xy + y^2$ 的全微分, 故通解为 $x^2 - xy + y^2 = C$. $\square$

$$6. y' = \frac{xy}{x^2 + y^2}.$$

解: 可变形为 $y' = \frac{1}{\frac{x}{y} + \frac{y}{x}}$.

令 $u = \frac{y}{x}$, 则 $\frac{dy}{dx} = \frac{du}{dx}x + u$.

代入得 $\frac{du}{dx}x + u = \frac{1}{\frac{1}{u} + u}$.

$$\text{即 } \frac{du}{dx}x = -\frac{u^3}{1 + u^2}$$

若 $u \neq 0$, 由分离变量法, 得 $\frac{1}{2u^2} - \ln u = \ln x + C$.

故通解为 $u = 0$ 或 $\frac{1}{2u^2} - \ln u = \ln x + C$. $\square$

$$7. y' = \frac{x}{x^2 + y^2}.$$

解: $xdx - (x^2 + y^2)dy = 0$

$$\frac{\partial M}{\partial y} = 0, \frac{\partial N}{\partial x} = -2x.$$

$$-\frac{1}{M}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = -2$$

故方程存在只与 y 有关的积分因子 $\mu(y) = e^{\int 2dy} = e^{-2y}$

两边同乘积分因子 $e^{-2y}xdx - e^{-2y}(x^2 + y^2)dy = 0$.

$e^{-2y}x$ 对 x 积分得 $\frac{1}{2}e^{-2y}x^2 + \phi(y)$

对 y 求导得 $-e^{-2y}x^2 + \phi'(y) = -e^{-2y}(x^2 + y^2)$.

解得 $\phi(y) = \frac{1}{2}(y^2 + y + \frac{1}{2})e^{-2y}$.

所以这是 $\frac{1}{2}e^{-2y}x^2 + \frac{1}{2}(y^2 + y + \frac{1}{2})e^{-2y}$ 的全微分

故解为 $\frac{1}{2}e^{-2y}x^2 + \frac{1}{2}(y^2 + y + \frac{1}{2})e^{-2y} = C$. $\square$

1.

$$y^{(6)} + y = 0$$

Answer 特征方程为 $t^6 + 1 = 0$, 故其特征根为

$$t_n = \exp\left(\frac{(2n+1)\pi}{6}i\right) \quad (n = 0, 1, 2, 3, 4, 5)$$

$$t_n = \frac{\sqrt{3}}{2} \pm \frac{1}{2}i, \pm i, -\frac{\sqrt{3}}{2} \pm \frac{1}{2}i$$

微分方程的通解为

$$y = \sum_{i=0}^5 c_i e^{t_i \cdot x}$$

$$y = e^{\frac{\sqrt{3}}{2}x} (C_1 \cos \frac{1}{2}x + C_1 \sin \frac{1}{2}x) + e^{-\frac{\sqrt{3}}{2}x} (C_3 \cos \frac{1}{2}x + C_4 \sin \frac{1}{2}x) + C_5 \cos x + C_6 \sin x$$

2.

$$y^{(4)} - 4y''' + 4y'' = 0$$

Answer 设 $u = y''$ 得

$$u'' - 4u' + 4u = 0$$

特征方程为 $t^2 - 4t + 4 = 0$, 故其特征根为 $x = 2$

$$y'' = u = (C_1 x + C_2)e^{2x} \quad y = \int \left(\int y'' dx \right) dx = (C_1 x + C_2)e^{2x} + C_3 x + C_4$$

3.

$$y^{(4)} + 2y''' + 2y'' = 3e^t + 2te^{-t} + e^{-t} \sin t$$

Answer 设 $u = y''$ 得该方程对应的齐次方程, 及其特征方程和特征根为

$$u'' + 2u' + 2u = 0 \quad t^2 + 2t + 2 = 0 \quad t = -1 \pm i$$

故齐次方程的通解为

$$y'' = u = e^{-t}(C_1 \cos t + C_2 \sin t) \quad y = \int \left(\int y'' dx \right) dx = e^{-t}(C_1 \cos t + C_2 \sin t) + C_3 x + C_4$$

容易观察, 改方程的特解 u_0 具有以下形式

$$u_0 = ae^t + bte^{-t} + e^{-t}f(t)$$

代入原方程可得 (Hint: 三个部分线性, 可以分开确定系数, 前两部分易得)

$$u_0 = \frac{3}{5}e^t + 2te^{-t} + e^{-t}f(t)$$

将上式代入原方程化简得到

$$f'' + f = \sin t$$

进而利用常数变易法得到该方程的通解

$$f(t) = C'_1 \cos t + C'_2 \sin t - \frac{1}{2}t \cos t$$

取 $C'_1 = C'_2 = 0$ 得到原方程的特解

$$y''_0 = u_0 = \frac{3}{5}e^t + 2te^{-t} - \frac{1}{2}te^{-t} \cos t$$

$$y_0 = \int \left(\int y''_0 dt \right) dt = \frac{3}{5}e^t + (4 + 2t)e^{-t} + \frac{1}{4}e^{-t}((t+1)\sin t + \cos t) + C_1 t + C_2$$

故原方程通解为

$$y = e^{-t}(C_1 \cos t + C_2 \sin t) + C_3 x + C_4 + \frac{3}{5}e^t + (4 + 2t)e^{-t} + \frac{1}{4}e^{-t}((t+1)\sin t + \cos t)$$

4.

$$y^{(6)} + y''' = t$$

Answer 设 $u = y'''$ 得该方程对应的齐次方程，及其特征方程和特征根为

$$u''' + u = 0 \quad t^3 + 1 = 0 \quad t = -1, \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$y''' = u = C_1 e^{-t} + e^{\frac{1}{2}t} \left(C_2 \cos \frac{\sqrt{3}}{2}t + C_3 \sin \frac{\sqrt{3}}{2}t \right)$$

$$y = C_1 e^{-t} + e^{\frac{1}{2}t} \left(C_2 \cos \frac{\sqrt{3}}{2}t + C_3 \sin \frac{\sqrt{3}}{2}t \right) + C_4 t^2 + C_5 t + C_6$$

容易看出，方程有特解

$$y_0 = \frac{1}{24}t^4$$

故原方程通解为

$$y = C_1 e^{-t} + e^{\frac{1}{2}t} \left(C_2 \cos \frac{\sqrt{3}}{2}t + C_3 \sin \frac{\sqrt{3}}{2}t \right) + C_4 t^2 + C_5 t + C_6 + \frac{1}{24}t^4$$

5.

$$y''' - y'' + y' - y = g(t) \text{ (解用积分的形式表示)}$$

Answer 该方程对应的齐次方程，及其特征方程和特征根为

$$y''' - y'' + y' - y = 0 \quad t^3 - t^2 + t - 1 = 0 \quad t = 1, \pm i$$

齐次方程的通解为

$$y = C_1 e^t + C_2 \cos t + C_3 \sin t$$

下面我们寻找特解，由常数变易法，记

$$y = C_1(t)y_1(t) + C_2(t)y_2(t) + C_3(t)y_3(t)$$

$C_1(t)$ 、 $C_2(t)$ 、 $C_3(t)$ 分别由下式给出

$$C_i(t) = \int \frac{g(s)\mathcal{W}_i(s)}{\mathcal{W}(s)} ds \quad (i = 1, 2, 3)$$

其中

$$\mathcal{W}(s) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^s & \cos t & \sin t \\ e^s & -\sin t & \cos t \\ e^s & -\cos t & -\sin t \end{vmatrix} = 2e^s$$

$$\mathcal{W}_1(s) = \begin{vmatrix} 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 1 & -\cos t & -\sin t \end{vmatrix} = 1$$

$$\mathcal{W}_2(s) = \begin{vmatrix} e^s & 0 & \sin t \\ e^s & 0 & \cos t \\ e^s & 1 & -\sin t \end{vmatrix} = e^s(-\cos s + \sin s)$$

$$\mathcal{W}_3(s) = \begin{vmatrix} e^s & \cos t & 0 \\ e^s & -\sin t & 0 \\ e^s & -\cos t & 1 \end{vmatrix} = -e^s(\cos s + \sin s)$$

$$C_1(t) = \frac{1}{2} \int g(s)e^{-s} ds \quad C_2(t) = \frac{1}{2} \int g(s)(\sin s - \cos s) ds \quad C_3(t) = -\frac{1}{2} \int g(s)(\sin s + \cos s) ds$$

综上所述，该方程的通解为

$$C_1 e^t + C_2 \cos t + C_3 \sin t + \frac{e^t}{2} \int g(s)e^{-s} ds + \frac{\cos t}{2} \int g(s)(\sin s - \cos s) ds - \frac{\sin t}{2} \int g(s)(\sin s + \cos s) ds$$